

ASHUTOSH MOHAN

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ACADEMIC PROJECTS

ReMusic – Desktop music player | C, C++

- Developed a desktop music player application in C++ using **Raylib**, implementing core playback features including play, pause, next/previous, track navigation, volume control, and a progress bar within an interactive GUI.
- Designed an efficient **song search system using a Trie data structure**, enabling fast lookup of tracks by title along with metadata display such as artists.
- Built a **playlist management system** using a circular doubly linked list, supporting dynamic addition and removal of songs and efficient traversal during playback.
- Implemented **Fisher–Yates (Martin Fielder) shuffle algorithm** for unbiased random playback and integrated **directory scanning** to load songs from local storage into the music library.

Tic-Tac-Toe & Minesweeper | Python

- Developed Tic-Tac-Toe and Minesweeper, enabling autonomous gameplay using algorithmic decision-making.
- Implemented the **Minimax algorithm** for Tic-Tac-Toe, allowing the program to evaluate game states and choose optimal moves for competitive play. And built a **logic-based inference system** for Minesweeper, enabling the program to deduce safe cells and mine locations based on revealed board information.
- Designed the games to support **both self play and human gameplay modes**, allowing users to play manually or observe the program solve the games.

YelpHall – Hall Booking & Review Platform | Node.js, Express, MongoDB, Bootstrap

- Developed a **full-stack web application** that enables users to discover, review, and book halls for events, creating a centralized platform for venue listings and user feedback.
- Built the backend using Node.js and Express, implementing RESTful routes to manage hall listings, bookings, and user-generated reviews with MongoDB for persistent data storage. And implemented secure user authentication using Passport.js and designed a responsive user interface with Bootstrap to support registration, login, and hall management features.
- Created an **interactive world map interface** where users can zoom, navigate, and click on registered halls to view their details, improving venue discovery and visual navigation across locations

EXPERIENCE

Vocational Trainee | BHEL, Haridwar

(May'25 – June'25)

- Collaborated on developing the prototype of a new material request system for BHEL computer department employees, with other trainees.
- Analyzed the existing system of wedding hall booking in BHEL township and developed a new prototype for a full stack website.

SKILLS

- C++, C, Python, SQL, MongoDB, Node.js, Express, React.js, Redux toolkit
- DSA, OOPS, Git, Raylib GUI, Bootstrap

EDUCATION

- Btech CSE | Amity university, Noida

(2022-2026)

CERTIFICATES

- Software Conceptual Design (NPTEL), Python for Data Science (NPTEL)